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神經網路與 Digital Twin 的資料中心即時溫度預測與冷卻控制

Real-time Temperature Prediction and Cooling Control for Data Centers Using Neural Networks and Digital Twin Technology.

1. Abstract

隨著雲端運算與人工智慧技術快速發展，資料中心的能源消耗與散熱問題日益嚴重。其中，冷卻系統佔據大量能耗，而傳統熱管理方法往往缺乏即時性、空間溫度感知能力與整體系統整合能力。現有研究多半僅專注於感測、預測或控制等單一模組，缺乏完整的端到端熱管理架構，導致 hotspot 偵測與動態冷卻控制效果受限。

本研究提出一個結合 Digital Twin 與 Generative Adversarial Network (GAN) 的即時熱管理框架，用於資料中心之空間溫度重建、溫度預測與動態冷卻控制。本研究假設 GAN 能夠比傳統 U-Net 方法更有效學習空間溫度分佈特徵，提升 hotspot detection 與即時 thermal prediction 的準確性。系統透過環境感測器收集資料，並利用 GAN 模型進行空間溫度場重建，預測結果同步至 Digital Twin 平台，以建立資料中心的虛擬模型並支援即時監測與決策。

本研究亦導入即時控制機制，根據系統狀態動態調整冷卻參數與工作負載分配，以降低能源消耗並提升 cooling efficiency。預期本研究可提升 thermal management 的即時性、空間感知能力與系統整合性，並提供未來智慧資料中心熱管理的重要參考。

This paper addresses the growing thermal management and energy consumption challenges in modern data centers. Traditional thermal management methods often lack real-time capability, spatial thermal awareness, and system-level integration, while many existing studies focus only on isolated modules such as sensing, prediction, or control.

Therefore, this paper proposes a real-time thermal management framework integrating Digital Twin technology and Generative Adversarial Networks (GANs) for spatial temperature reconstruction, temperature prediction, and dynamic cooling control in data centers. This paper assumes that GANs can learn spatial temperature distributions more effectively than traditional U-Net methods,

improving hotspot detection and real-time thermal prediction accuracy.

Environmental sensor data are used for thermal prediction and synchronize with a Digital Twin platform for real-time monitoring and decision-making. In addition, an adaptive control mechanism dynamically adjusts cooling parameters and workload distribution to improve cooling efficiency and reduce energy consumption.

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